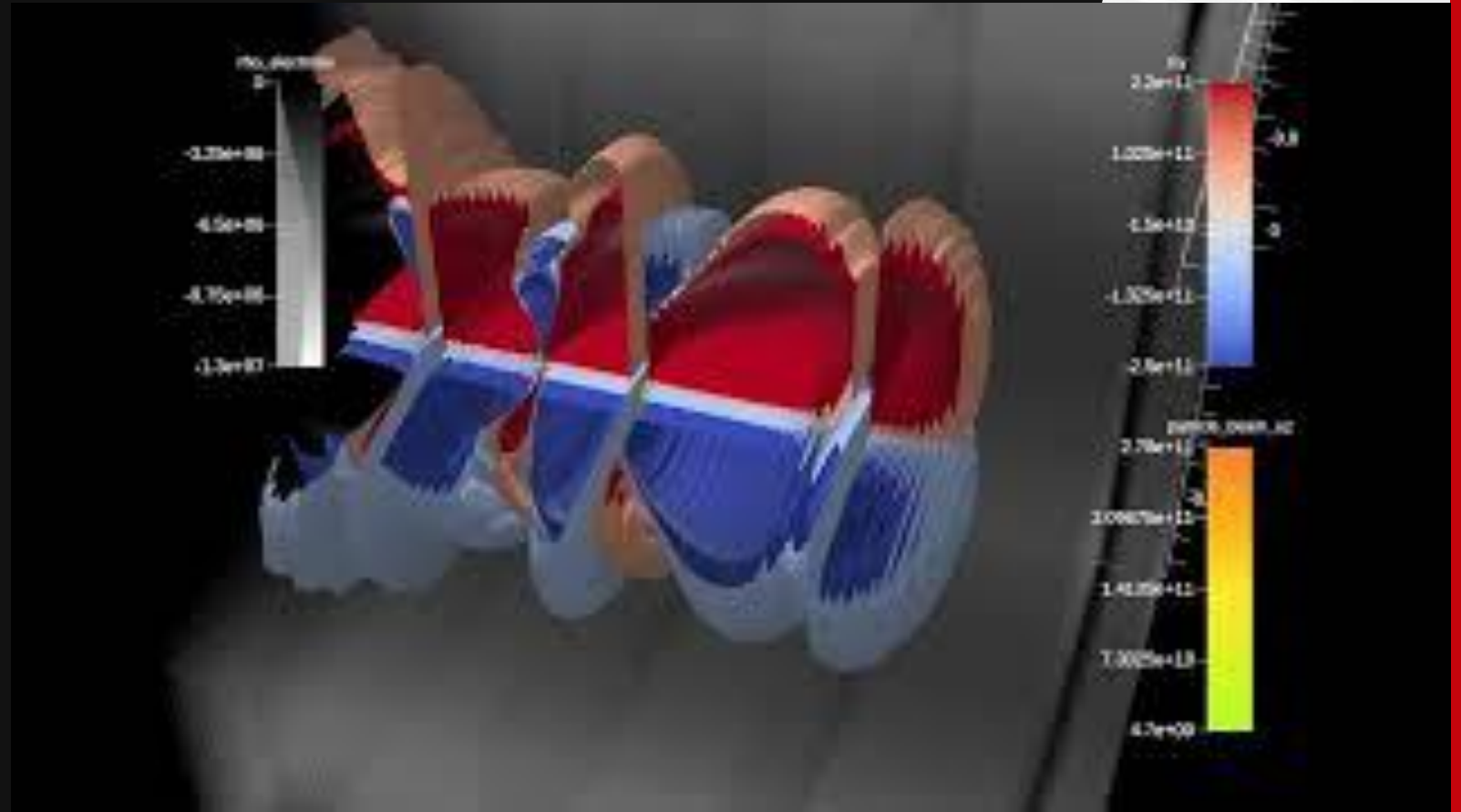


Plasma Wakefield Acceleration

Jason Xu and Avery Kahar



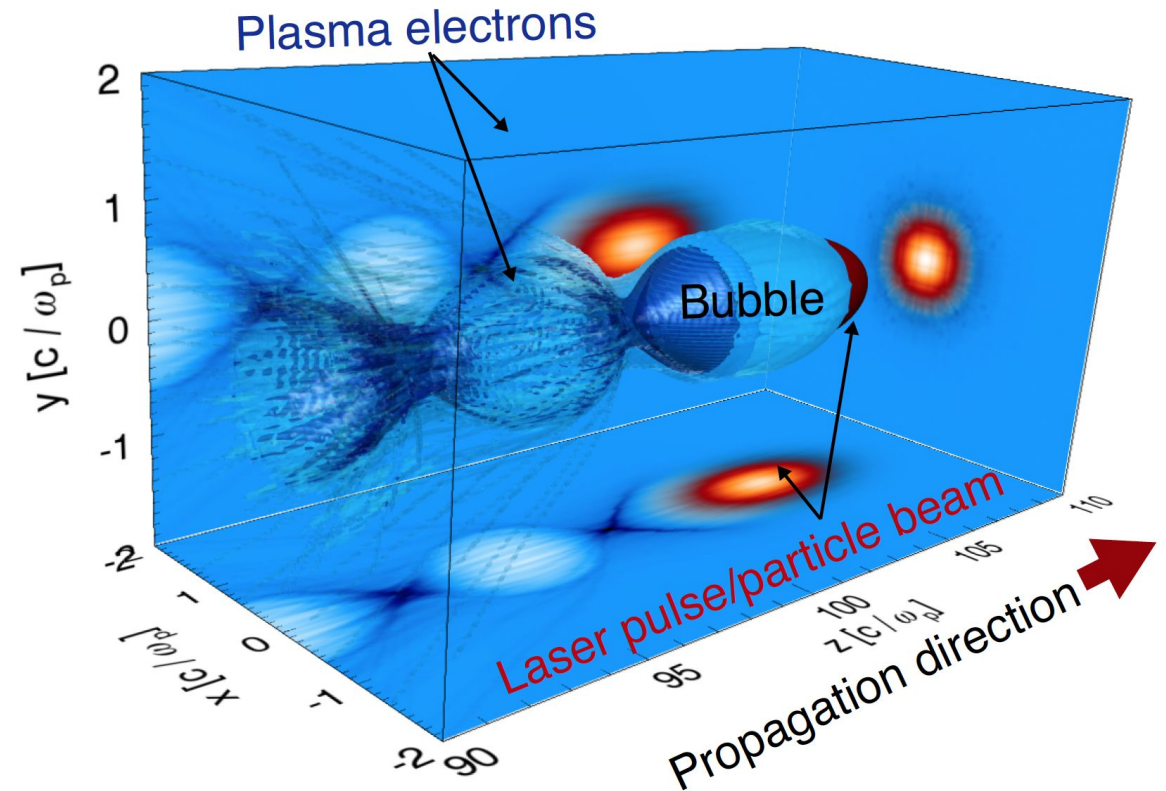
Contents

- ❖ Introduction
- ❖ 1D Toy Model Simulations
- ❖ WarpX Simulations

Main takeaways: Formulation, qualitative behavior of beams and field, and beam optimization/validation of 1D formulation

Introduction

- PWFA has much higher accelerating gradients vs traditional RF cavities
- Shoot a drive beam through plasma to create a wakefield.
- A witness beam is carefully placed to gain energy in the wake.
- Energy is transferred from drive beam to witness beam
- <https://www.youtube.com/watch?v=L75CfsiHJRg>



Toy Model

- ❖ 1 Dimension
- ❖ Cold-fluid
- ❖ Immobile Ions
- ❖ Linear Response
- ❖ Electrostatic reduction
- ❖ Quasi-static (more on this later)
- ❖ Non-relativistic
- ❖ No magnetic fields
- ❖ Valid because same plasma frequency

Continuity: $\frac{\partial n_1}{\partial t} + n_0 \frac{\partial v_1}{\partial z} = 0$

Momentum: $m_e \frac{\partial v_1}{\partial t} = -eE$

Poisson: $\frac{\partial E}{\partial z} = -\frac{e}{\epsilon_0} (n_1 + n_b)$

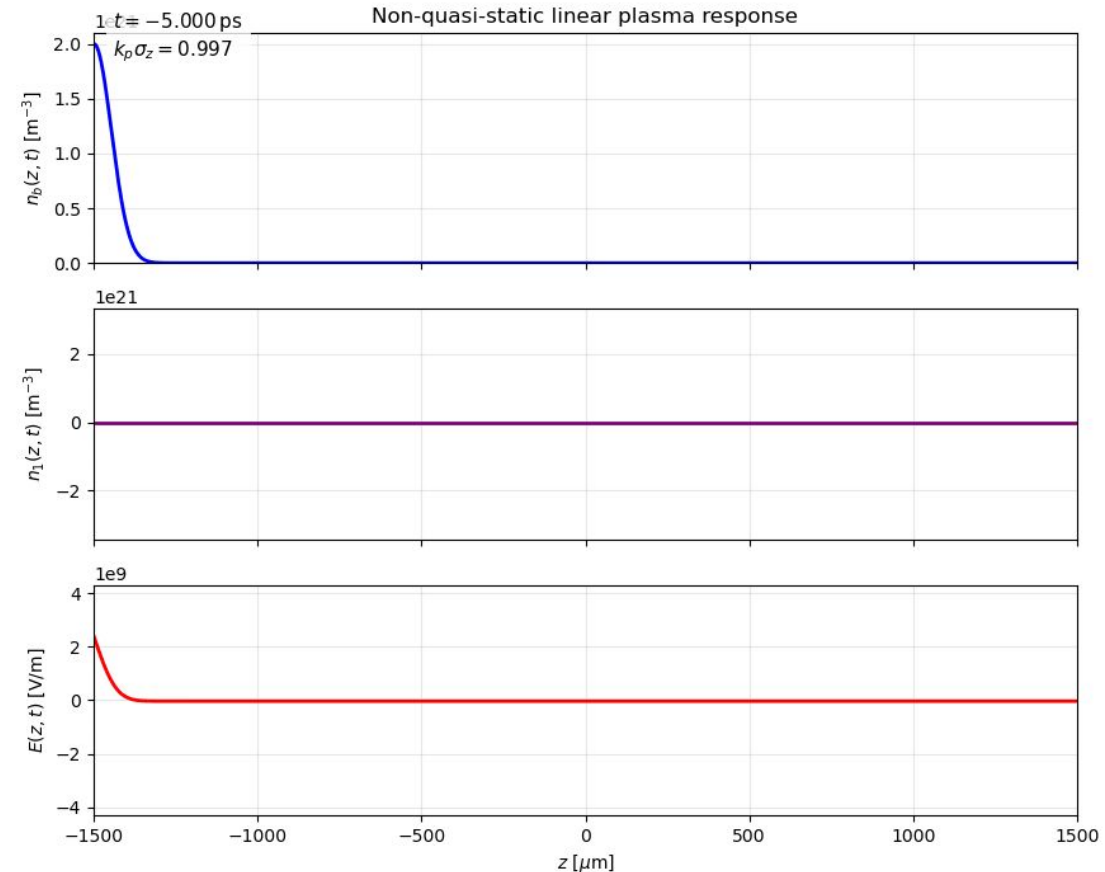
$$\frac{\partial}{\partial t} \left(\frac{\partial n_1}{\partial t} + n_0 \frac{\partial v_1}{\partial z} \right) = 0$$

$$\frac{\partial^2 n_1}{\partial t^2} - \frac{en_0}{m_e} \frac{\partial E}{\partial z} = 0$$

$$\frac{\partial^2 n_1}{\partial t^2} + \underbrace{\frac{n_0 e^2}{m_e \epsilon_0}}_{:= \omega_p^2} (n_1 + n_b) = 0$$

$$\frac{\partial^2 n_1}{\partial t^2} + \omega_p^2 n_1 = -\omega_p^2 n_b$$

Numerical Solution



$$\xi = z - ct, \quad \frac{\partial}{\partial t} \rightarrow -c \frac{d}{d\xi}, \quad \frac{\partial}{\partial z} \rightarrow \frac{d}{d\xi}$$

$$\frac{d^2 n_1}{d\xi^2} + \underbrace{\frac{\omega_p^2}{c^2}}_{:= k_p^2} n_1 = -k_p^2 n_b$$

$$\frac{dE}{d\xi} = -\frac{e}{\epsilon_0} (n_1 + n_b)$$

$$n_1 = -\frac{\epsilon_0}{e} \frac{dE}{d\xi} - n_b$$

$$-\frac{\epsilon_0}{e} \frac{d^3 E}{d\xi^3} - \frac{d^2 n_b}{d\xi^2} - k_p^2 \frac{\epsilon_0}{e} \frac{dE}{d\xi} - k_p^2 n_b = -k_p^2 n_b$$

$$\frac{d}{d\xi} \left(\frac{d^2 E}{d\xi^2} + k_p^2 E \right) = -\frac{e}{\epsilon_0} \frac{d^2 n_b}{d\xi^2}$$

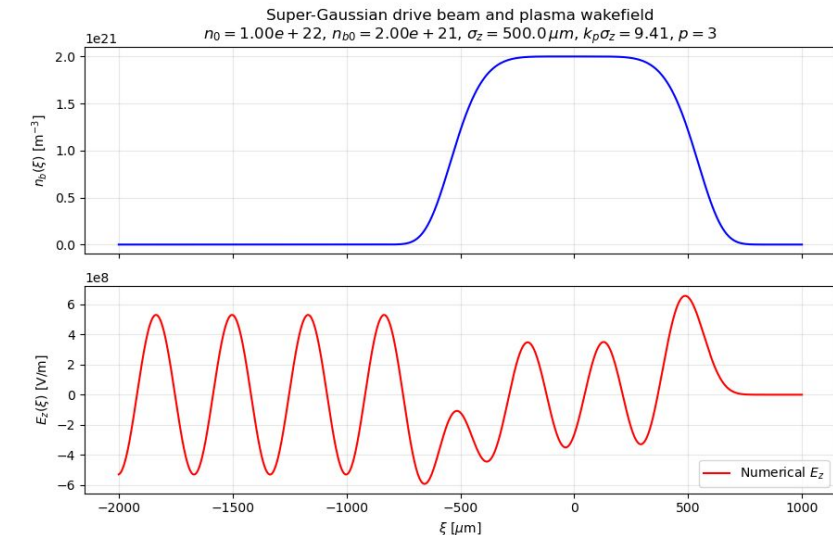
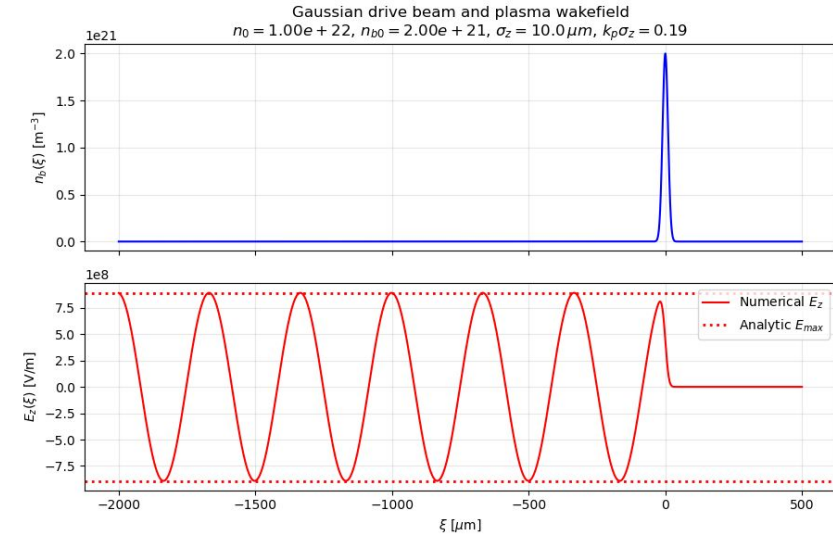


Assumptions:

- $n_{\text{beam}}(z, t) = n_{\text{beam}}(\xi)$ (Quasi-static assumption). Everything is assumed to be “still” in the beams frame. ξ is distance in that frame.
- $v_{\text{beam}} = c$. This is justified because electrons require low energy to accelerate to $\sim c$.
- $n_{\text{beam}}(\xi)$ does not deform as it travels through the plasma. So we are ignoring feedback and beam-loading effects.

We show solutions for $E(\xi)$ on the right. It is the spacial derivative of $n_{\text{beam}}(\xi)$ that acts as a forcing term for $E(\xi)$.

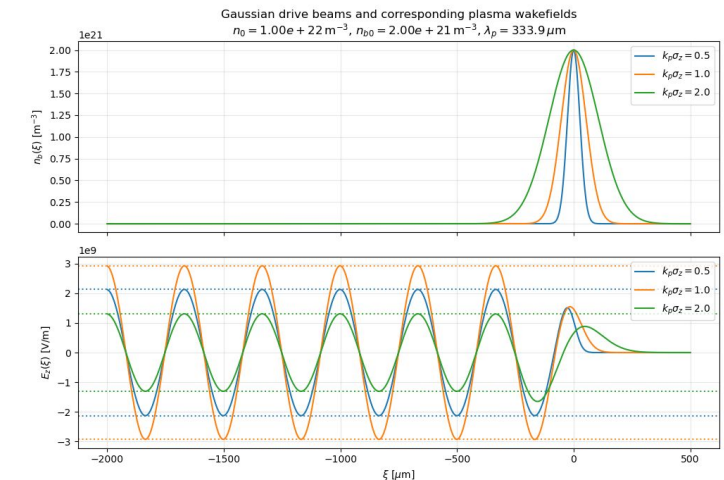
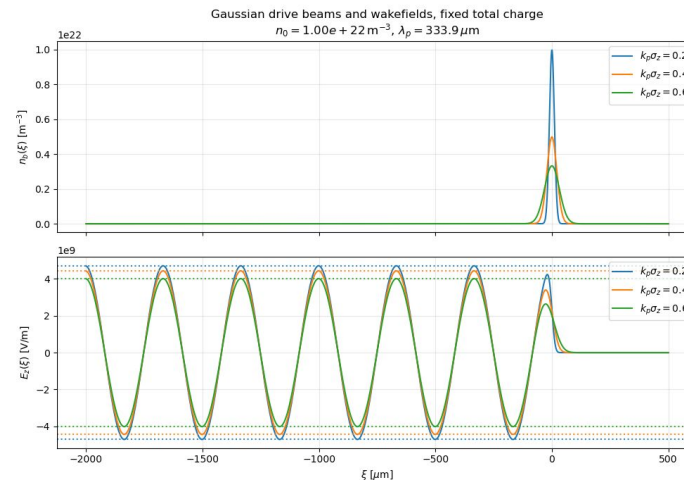
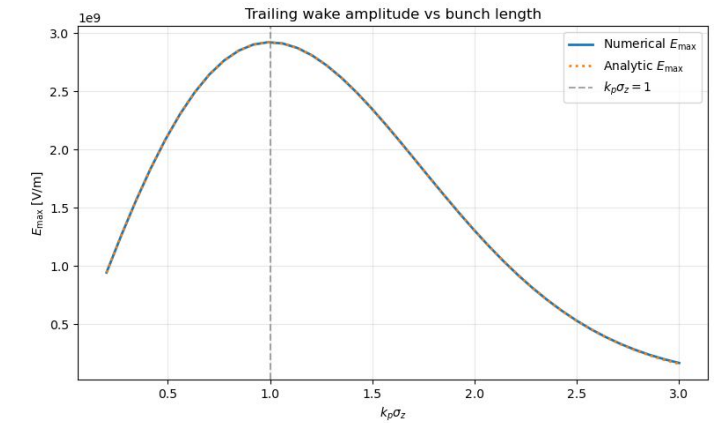
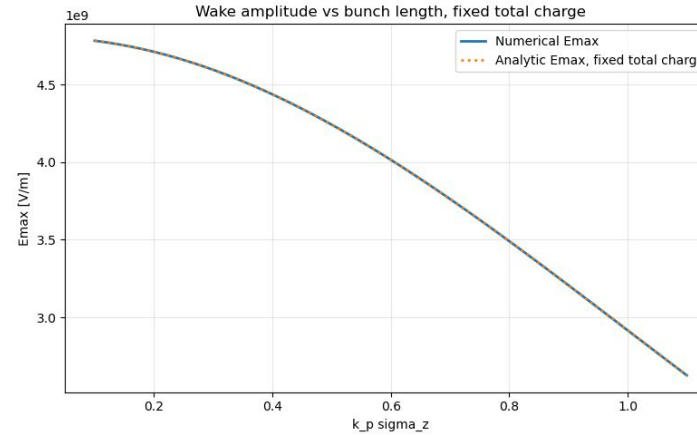
$$\frac{d^2 E}{d\xi^2} + k_p^2 E = -\frac{e}{\epsilon_0} \frac{dn_b}{d\xi}$$



- Compare:
 - width of drive beam
 - amplitude of the wake
- These relations can be derived analytically (dotted line)
- The energy gain is:

$$\frac{\Delta \mathcal{E}}{L} = \frac{q}{L} \int_0^L E_z(z) dz = q E_{\text{eff}} \approx 1 \text{ GeV/m}$$

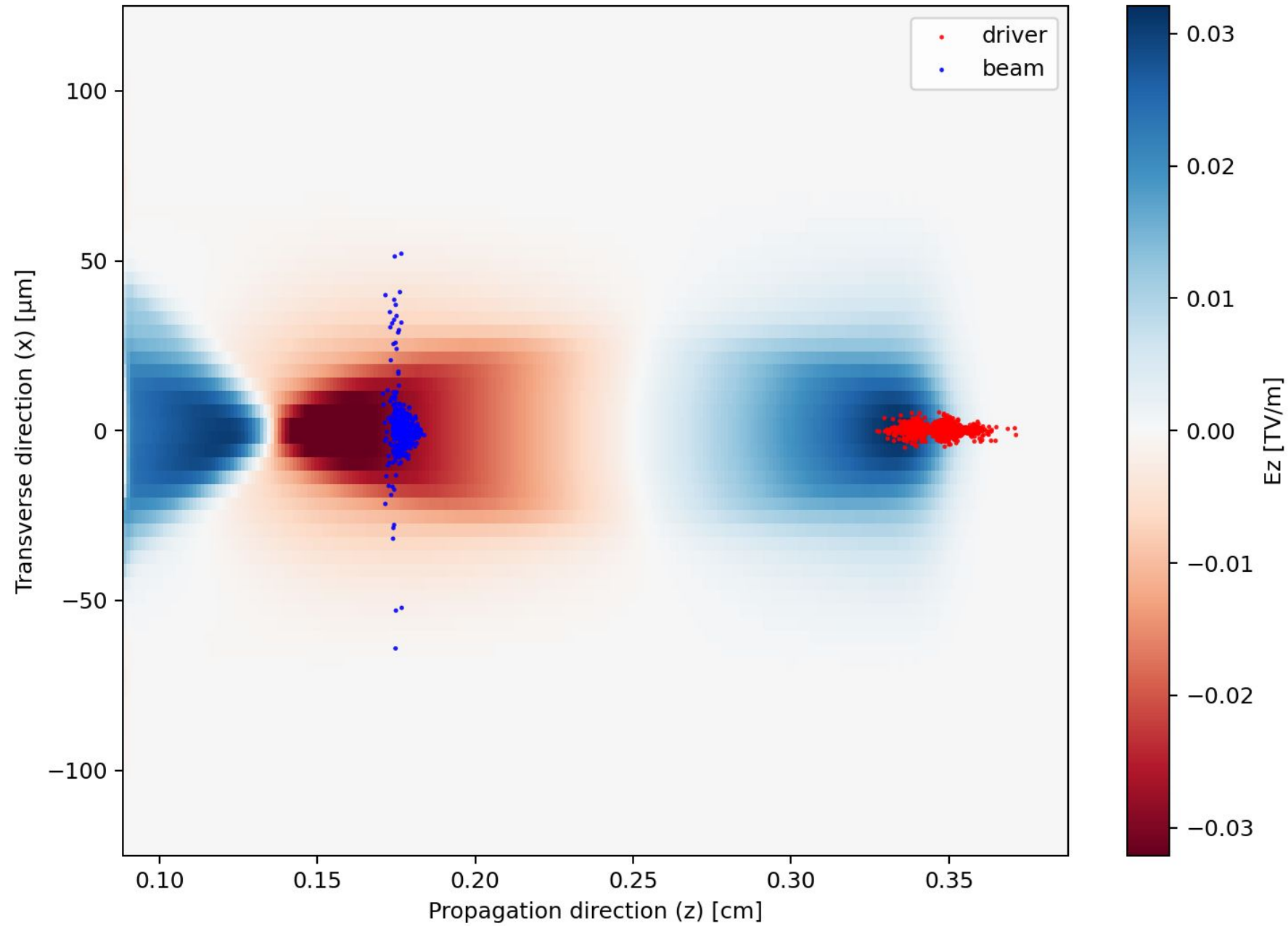
- ~10x stronger than a traditional RF cavity.
- agrees with experiments
- Witness beam usually will not be placed at E_{max} since accelerator physicists also want longitudinal focusing.
- Witness beam is placed in first bubble, since later bubbles are chaotic (characterized experimentally)

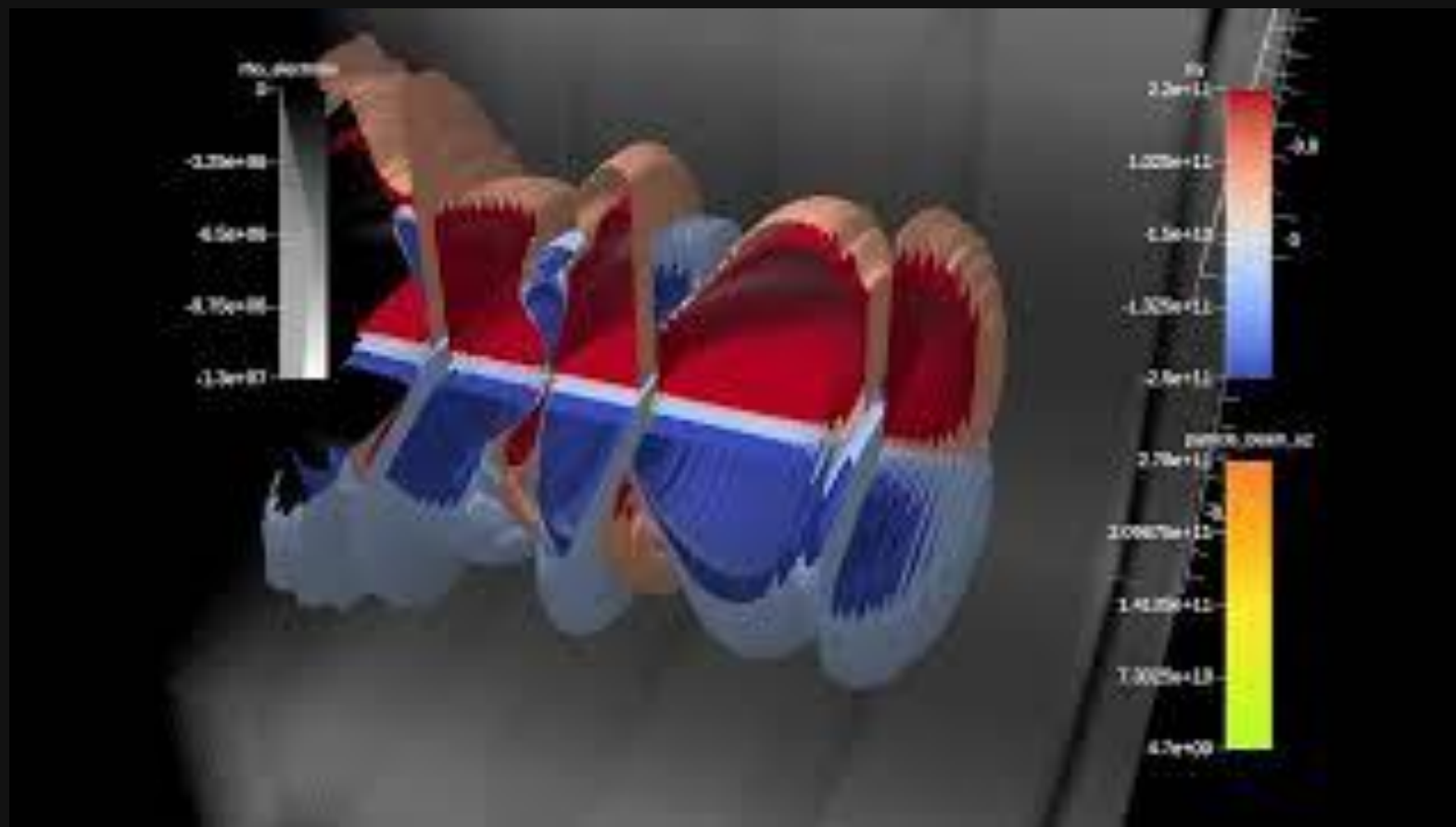


WarpX Simulations

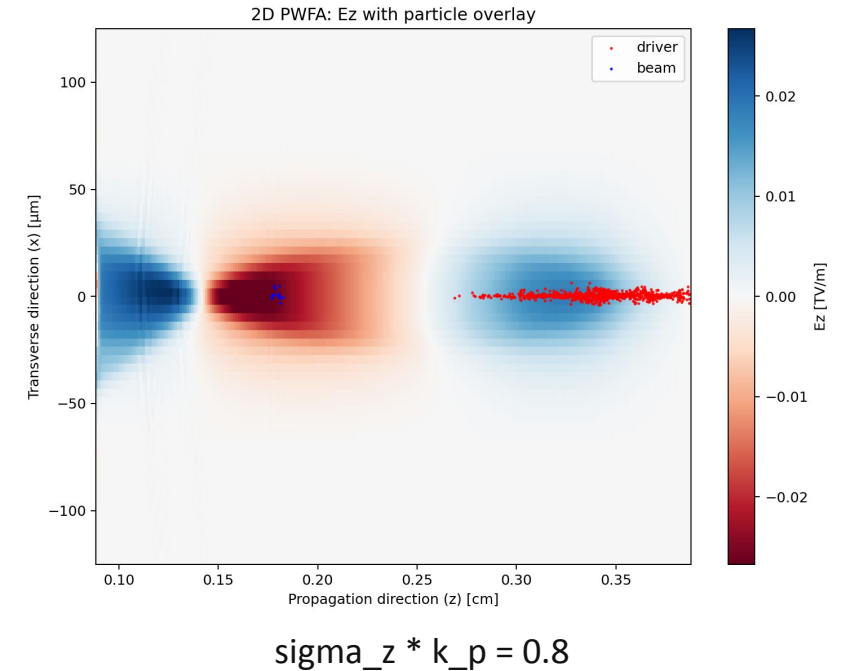
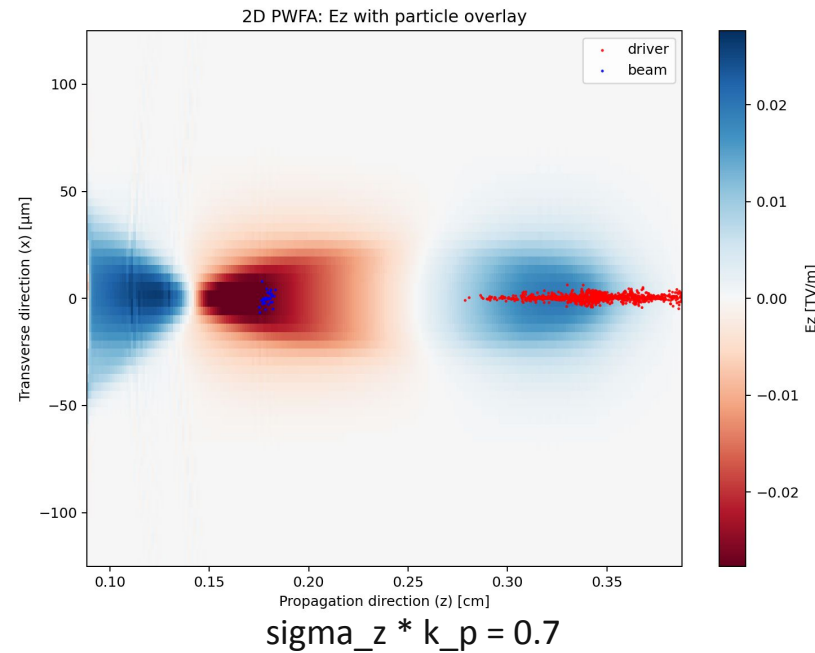
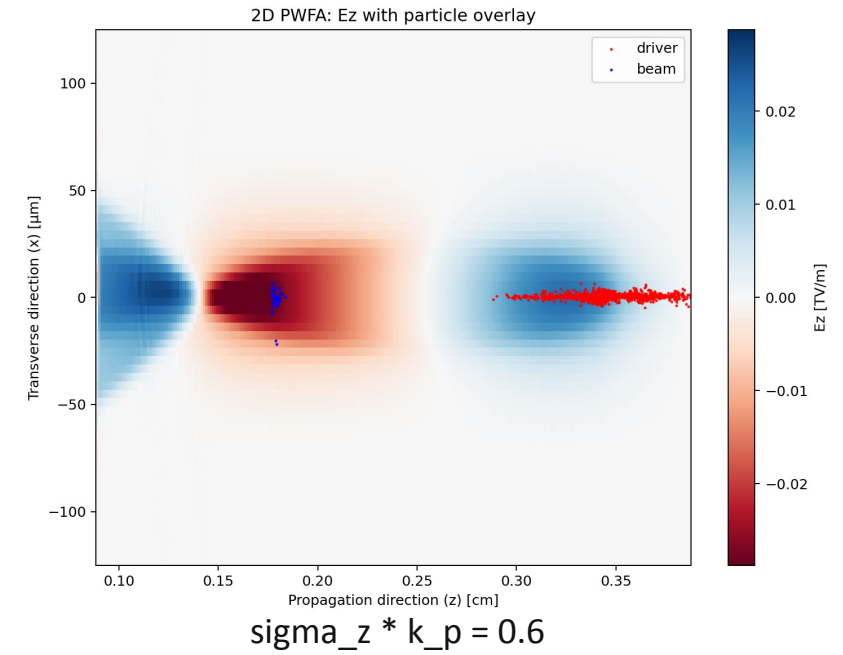
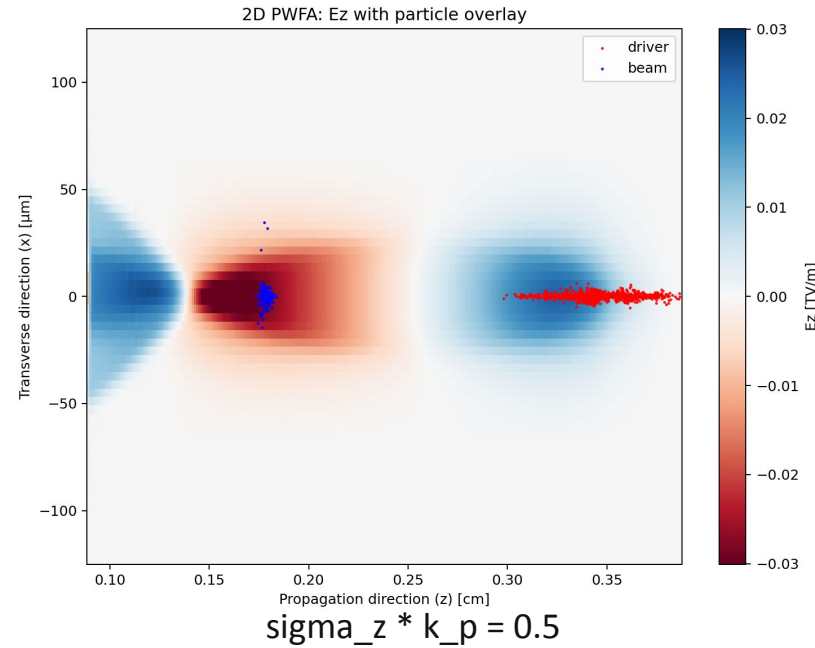


2D PWFA: Ez with particle overlay

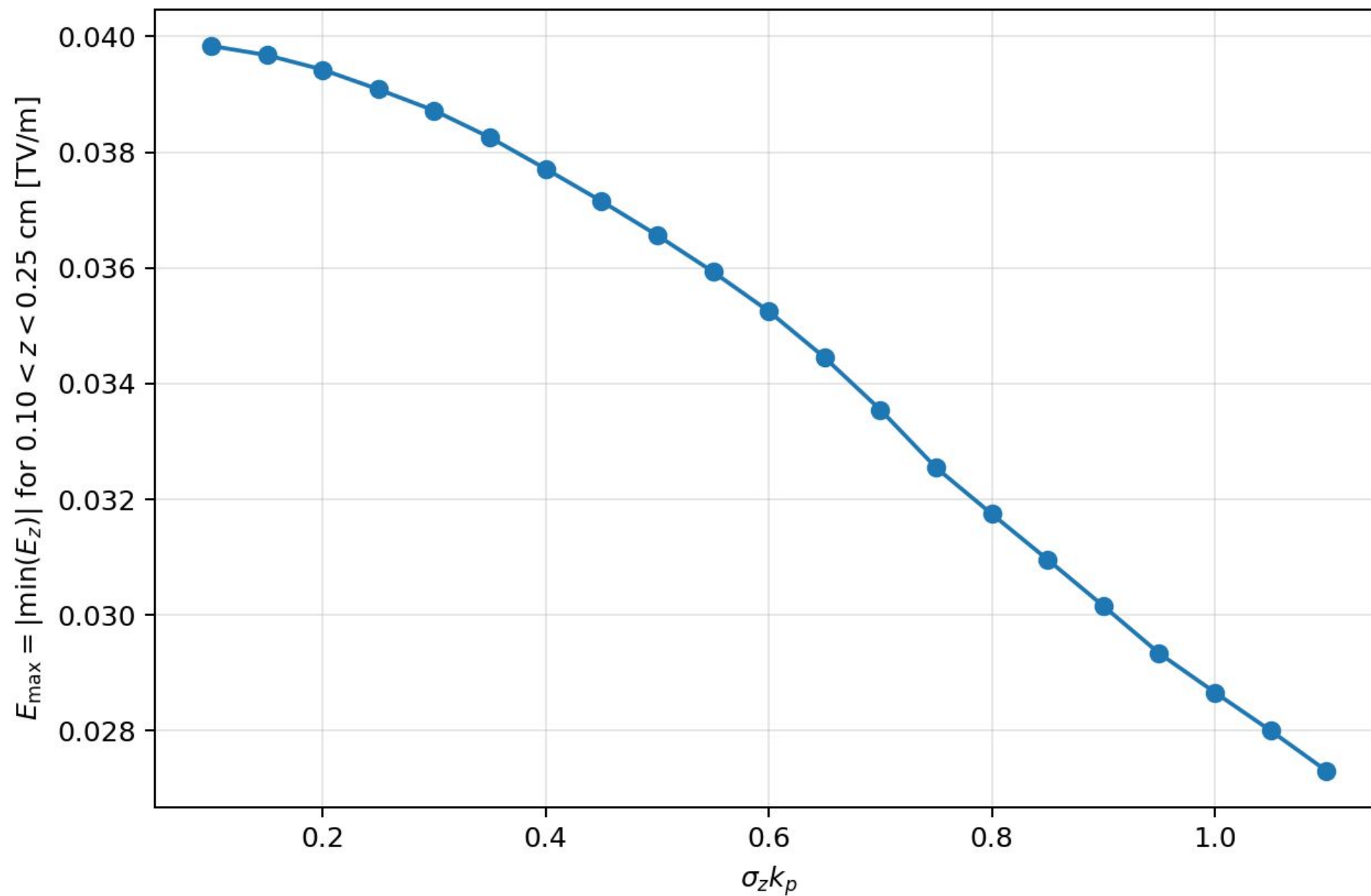


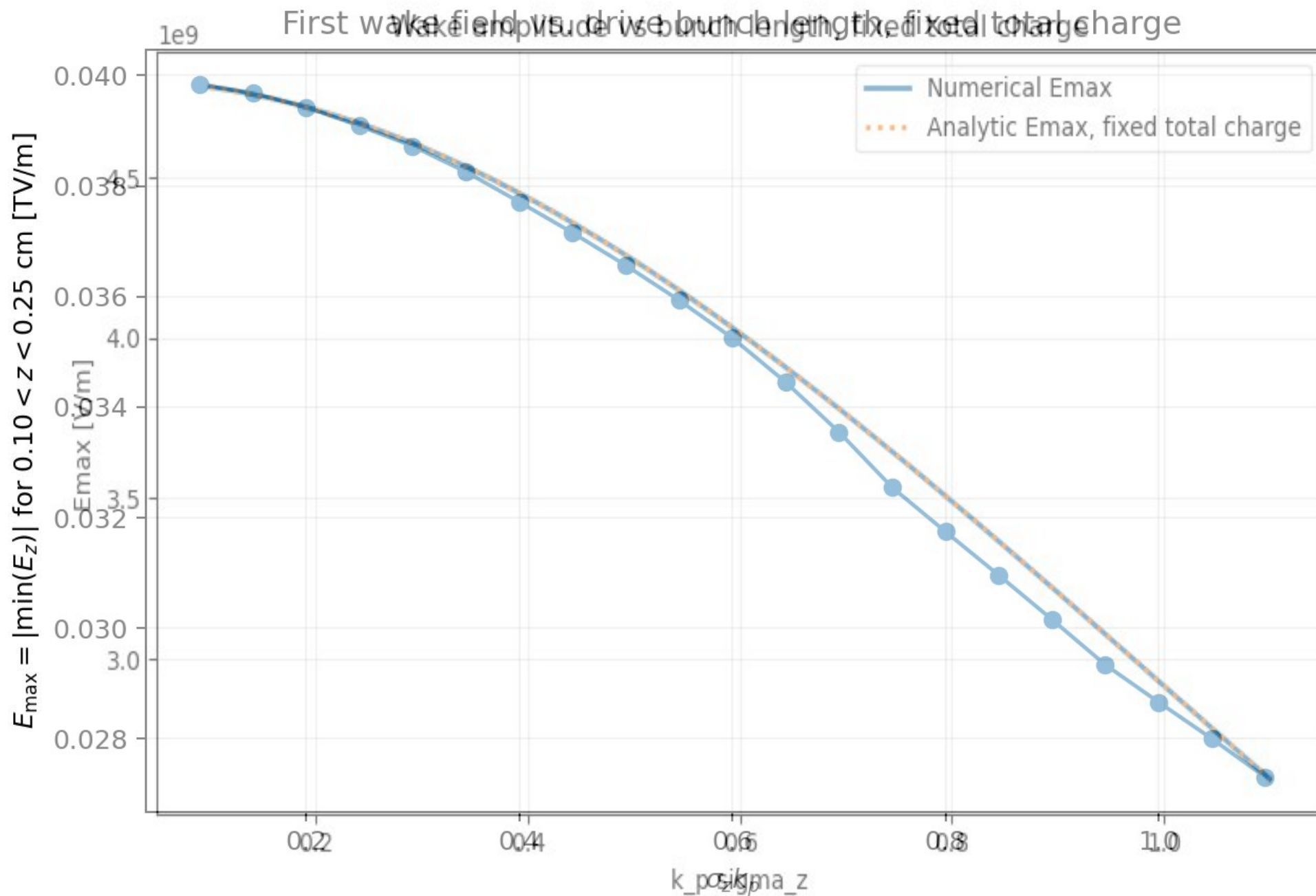


- Wakes are same magnitude
- Same relation between drive beam size and wake strength
- Wakefields are ~30 GeV



First wake field vs. drive bunch length, fixed total charge





Awesome

